

Chip (in)dependence

INDIA HAS LONG PASSED THE TODDLER STAGE, LEARNING HOW TO WALK BEFORE WE RUN IN THE FIRST FEW DECADES AFTER INDEPENDENCE, and even passed through the awkward teenage phase where we matured and found our calling. It's hard to think on an empty stomach, so India needed a green revolution to make us totally self-sufficient when it came to feeding our masses. That happened between the 1960s and 1970s, and once we were properly fed, and could dream bigger, we opened up to the world.

However, it was in the 1990s that we dove into the mad rush that was the technology boom. Digit itself was born in 2001, and wow, what a journey the tech revolution in India has been! The mobile revolution especially has been something the world marvels. I've been privileged enough to own a mobile phone from the 90s where a mobile phone call would cost ₹16 per minute! No, it wasn't because I was rich enough to afford that, I was poor. It was because I was too stupidly in love with technology to miss out on it, and I'd rather experience it than eat all three meals a day. And I wasn't alone – thousands of people like me were FoMo-ing way before the acronym was coined.

Digit was a haven for people like me, and that hasn't changed at all actually. We still attract the technology lovers, all of whom could work elsewhere and earn more, but we all choose to run the gauntlet that is technology journalism because we love the subject matter.

Since 2001 Digit has been eagerly tracking the progress of every aspect of technology growth in India, egging it on, waiting for the tech virus to reach critical mass, and boy has it been fun! I think it's safe to say that India is now hooked on tech. Thanks to a million innovators, startups, established companies and of course, a billion people who are fascinated by technology, India is charting her own course in the technology ocean, and solving uniquely Indian problems while sailing those turbulent seas.

What we are lacking is something akin to the green revolution, which made us self-reliant and self-sufficient when it came to food security. We need a chip revolution to do the same for technology in India, especially when it comes to manufacturing. In December 2021 the government announced a \$10 billion incentive plan for chip manufacturers looking to set up shop in India, and this is a great move. I think it's

at least ten times too small, and it also needs sustained support. And my opinion is insignificant, of course, and it's just another opinion in the sea of opinions, none of whom seem to agree.

Many believe that India should not even waste time and money on trying to become technologically self-sufficient, or become a chip manufacturing hub, because the path is hard, and expensive, and littered with failures. The example of China is often thrown about – China went from manufacturing 0% of global chips to ~25% in a matter of decades. However, people are quick to point out that China themselves only manufacture lower grade, larger fab and lower complexity chips. They're at least 5 to 7 years behind the leaders of the industry such as TSMC (Taiwan Semiconductor Manufacturing Company). That alone is considered a big enough reason for why China wants to annex Taiwan. After all, TSMC produces nearly 90% of the world's advanced semiconductors. In terms of all semiconductors produced, China accounts for 24 per cent, Taiwan 21 per cent, South Korea 19 per cent, Japan 13 per cent, US 10 per cent, and the EU 8 per cent.

Despite all of the might of Chinese manufacturing, if China still cannot break into the top tier of advanced chip manufacturing, what hope does India have of becoming a chip manufacturing hub?

India and China are often looked at as similar because they both have big populations. Nothing is further from the truth. India is an open market, and it has good standing on the international stage. Western countries do not trust China, and this has impeded their potential growth. With 90 per cent of advanced chip manufacturing from Taiwan in danger of being taken over by China, a reliable and trustworthy alternative is needed. While South Korea and Japan, the US and EU are all capable of ramping up production it will come at higher running costs.

India's future as a chip manufacturer is far from dead. It might just need a bigger shot in the arm, but it will still see slow but steady growth. The sheer wealth of knowledge and expertise that will be developed and passed on within the country will be priceless. I'm hopeful that before we celebrate a centenary of independence, we will celebrate our independence from chip-dependence. Just make sure you do your bit and buy Indian-made chips when they come... 



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